Bharatram Vaidyanathan had spent two decades immersed in the fast-paced world of IT. But as the years passed, he found himself growing increasingly frustrated and weary of the relentless pressure and stress that came with his job. He yearned to return to his roots in Michaelpatti, his native village. So he quit his high-paying IT job and headed back to his homeland, seeking a life that was more in harmony with nature.

When Bharat returned to Michaelpatti, he was greeted by acres of uncultivated land surrounding his own plot. This sight inspired him to expand his land holdings, purchasing an additional 40 acres. However, this decision came with its own set of challenges.

The first problem Bharat faced was a severe shortage of labour. The younger generation had mostly migrated to cities in search of better job opportunities, leaving a scarcity of skilled farmworkers in the village. Moreover, water wastage was rampant due to outdated irrigation methods, and cattle encroachment was damaging his crops. Weeds were also taking over the fields, competing with his crops for valuable nutrients.

Determined to turn things around and make a success of his agricultural venture, Bharat began researching modern farming techniques. It wasn't long before he stumbled upon the concept of IoT (Internet of Things) in agriculture. He realised that this technology could be the solution to his problems.

Bharat started by implementing IoT sensors in his fields. Soil moisture sensors were installed to monitor the exact moisture levels in the soil, allowing him to optimise irrigation and reduce water wastage. He also installed cameras and motion sensors to deter cattle from entering his fields.

To address the labour shortage, Bharat invested in agricultural drones equipped with cameras and sensors. These drones could monitor the crops, identify weed-infested areas, and even assist in planting seeds and applying fertilisers.

He then set up a smart irrigation system that could be controlled remotely from his smartphone. With weather data from IoT-connected weather stations, he could make informed decisions about when and how much to irrigate, further conserving water and improving crop yields.

As time passed, Bharat's farm started to flourish. The IoT technology he had embraced had transformed the way he managed his land. He no longer had to worry about water wastage, cattle intrusion, or labour shortages. His fields were weed-free, and his crops were thriving.

Word of Bharat's success soon spread throughout the village, inspiring other farmers to adopt IoT in their farming practices. Together, they were not only reaping the benefits of increased productivity but also contributing to the sustainable and economic development of their taluk.

IoT in agriculture

IoT has significant applications in agriculture, often referred to as 'agri IoT' or 'smart farming'. IoT technology enables farmers to collect and analyse real-time data from their farms, making agriculture more efficient and adaptable to changing conditions.

IoT in agriculture has the potential to revolutionise the industry by increasing productivity, reducing costs, and promoting sustainability. It empowers farmers with valuable data-driven insights and tools to make more informed decisions.

Here are some key aspects of IoT in agriculture.

Precision agriculture: IoT sensors, such as soil moisture sensors, weather stations, and GPS-enabled equipment, allow farmers to monitor and control various factors like soil conditions, temperature, humidity, and crop health with high precision. This data helps optimise resource usage, reduce waste, and increase crop yields.

Crop monitoring: IoT devices can be used to monitor crop growth and health. Drones equipped with cameras and sensors can capture images of fields and provide valuable insights into plant health, pest infestations, and crop stress. Farmers can then take targeted actions to address these issues.

Livestock management: IoT devices like RFID tags and wearable sensors can be used to monitor the health and location of livestock. Farmers can track individual animals, monitor their vital signs, and receive alerts in case of illness or distress.

Smart irrigation: IoT-based irrigation systems can be programmed to deliver water only when and where it's needed. Soil moisture sensors can detect moisture levels and trigger irrigation systems accordingly, reducing water wastage and optimising crop hydration.

Weather forecasting: Access to real-time weather data through IoT helps farmers make informed decisions about planting, harvesting, and pest control. It allows them to mitigate the risks associated with adverse weather conditions.

Supply chain management: IoT can improve the tracking and traceability of agricultural products throughout the supply chain. Sensors and RFID tags can be used to monitor the temperature and humidity of crops during storage and transportation, ensuring product quality.

Smart greenhouses: IoT sensors and actuators can be used to control temperature, humidity, and lighting in greenhouses automatically. This optimises growing conditions for crops, extending the growing season and improving yield.

Data analytics: IoT-generated data can be analysed using advanced analytics and machine learning algorithms to provide actionable